- Installing docker:

```
aiysha@aiysha-VMware-Virtual-Platform:~$ sudo systemctl status docker
● docker.service - Docker Application Container Engine
     Loaded: loaded (/usr/lib/systemd/system/docker.service; enabled; preset: enabled)
     Active: active (running) since Thu 2026-02-26 19:53:58 EET; 2min 44s ago
 Invocation: 96acf8cfeb5a40788de95b8c40492773
TriggeredBy: ● docker.socket
       Docs: https://docs.docker.com
   Main PID: 5464 (dockerd)
      Tasks: 9
     Memory: 26.1M (peak: 27M)
        CPU: 230ms
     CGroup: /system.slice/docker.service
             └─5464 /usr/bin/dockerd -H fd:// --containerd=/run/containerd/containerd.sock
```

- Test:

```
aiysha@aiysha-VMware-Virtual-Platform:~$ sudo docker run hello-world
Unable to find image 'hello-world:latest' locally
latest: Pulling from library/hello-world
17eec7bbc9d7: Pull complete

ea52d2000f90: Download complete

Digest: sha256:ef54e839ef541993b4e87f25e752f7cf4238fa55f017957c2eb44077083d7a6a
Status: Downloaded newer image for hello-world:latest

Hello from Docker!
This message shows that your installation appears to be working correctly.
```

- Installing nginx:alpine and exploring "docker run" command and run nginx: alpine image on the machine in the background and specify the name of the container to name-ITI:

```
aiysha@aiysha-VMware-Virtual-Platform:~$ sudo docker images

                                                  i Info →   U   In Use
IMAGE                 ID             DISK USAGE   CONTENT SIZE   EXTRA
hello-world:latest    ef54e839ef54       25.9kB         9.52kB    U
nginx:alpine          1d13701a5f9f       93.4MB         26.9MB
aiysha@aiysha-VMware-Virtual-Platform:~$ sudo docker run -d --name name-ITI -p 8080:80 nginx:alpine
4b686eeb9659d5df831e1b20980c869246ca1f6a0068e0310978978be2fa3251
```

→hint: -p 8080 means port forwarding was used here to forward traffic from host port 8080 to container's port 80

- Installing and running apache container with memory limit 70 MB and 1 core of CPU:

```
aiysha@aiysha-VMware-Virtual-Platform:~$ sudo docker pull httpd
Using default tag: latest
latest: Pulling from library/httpd
47a6bbb8d6d8: Pull complete
206356c42440: Pull complete
7079f4501db6: Pull complete
242bb693a1d3: Pull complete
4f4fb700ef54: Pull complete
80414dfb8f15: Pull complete
ea9cd2f459bd: Download complete
44fabeaa234c: Download complete
Digest: sha256:96b1e8f69ee3adde956e819f7a7c3e706edef7ad88a26a491734015e5c595333
Status: Downloaded newer image for httpd:latest
docker.io/library/httpd:latest
aiysha@aiysha-VMware-Virtual-Platform:~$ sudo docker run -d --name apache-ITI --memory="70m" --cpus="1" -p 8081:80 httpd
[sudo: authenticate] Password:
1cd6a68cd2da420ca5764ffd6ad63d5780a78708384554233cfb94e78b48d4f3
aiysha@aiysha-VMware-Virtual-Platform:~$
```

→ for apache, 8081 was used in port forwarding

- Running:

```
aiysha@aiysha-VMware-Virtual-Platform:~$ sudo docker ps
CONTAINER ID   IMAGE          COMMAND                  CREATED          STATUS          PORTS                                       NAMES
1cd6a68cd2da   httpd          "httpd-foreground"       7 minutes ago    Up 7 minutes    0.0.0.0:8081->80/tcp, [::]:8081->80/tcp     apache-ITI
4b686eeb9659   nginx:alpine   "/docker-entrypoint.…"   35 minutes ago   Up 35 minutes   0.0.0.0:8080->80/tcp, [::]:8080->80/tcp     name-ITI
aiysha@aiysha-VMware-Virtual-Platform:~$
```

- After creating the containers, the path of the cgroups created for this container and memory limit file and value:

```
aiysha@aiysha-VMware-Virtual-Platform:~$ ls /sys/fs/cgroup/system.slice/docker-1cd6a68cd2da420ca5764ffd6ad63d5780a78708384554233cfb94e78b48d4f3.scope/
cgroup.controllers      cpu.pressure                    dmem.low                hugetlb.2MB.rsvd.current  memory.oom.group        misc.events.local
cgroup.events           cpu.stat                        dmem.max                hugetlb.2MB.rsvd.max      memory.peak              misc.max
cgroup.freeze           cpu.stat.local                  dmem.min                io.max                    memory.pressure          misc.peak
cgroup.kill             cpu.uclamp.max                  hugetlb.1GB.current     io.pressure               memory.reclaim           pids.current
cgroup.max.depth        cpu.uclamp.min                  hugetlb.1GB.events      io.prio.class             memory.stat              pids.events
cgroup.max.descendants  cpu.weight                      hugetlb.1GB.events.local io.stat                  memory.swap.current      pids.events.local
cgroup.pressure         cpu.weight.nice                 hugetlb.1GB.max         io.weight                 memory.swap.events       pids.max
cgroup.procs            cpuset.cpus                     hugetlb.1GB.numa_stat   memory.current            memory.swap.high         pids.peak
cgroup.stat             cpuset.cpus.effective           hugetlb.1GB.rsvd.current memory.events            memory.swap.max          rdma.current
cgroup.subtree_control  cpuset.cpus.exclusive           hugetlb.1GB.rsvd.max    memory.events.local       memory.swap.peak         rdma.max
cgroup.threads          cpuset.cpus.exclusive.effective hugetlb.2MB.current     memory.high               memory.zswap.current
cgroup.type             cpuset.cpus.partition           hugetlb.2MB.events      memory.low                memory.zswap.max
cpu.idle                cpuset.mems                     hugetlb.2MB.events.local memory.max               memory.zswap.writeback
cpu.max                 cpuset.mems.effective           hugetlb.2MB.max         memory.min                misc.current
cpu.max.burst           dmem.current                    hugetlb.2MB.numa_stat   memory.numa_stat          misc.events
aiysha@aiysha-VMware-Virtual-Platform:~$ ls /sys/fs/cgroup/system.slice/docker-1cd6a68cd2da420ca5764ffd6ad63d5780a78708384554233cfb94e78b48d4f3.scope/cpu
.max
/sys/fs/cgroup/system.slice/docker-1cd6a68cd2da420ca5764ffd6ad63d5780a78708384554233cfb94e78b48d4f3.scope/cpu.max
aiysha@aiysha-VMware-Virtual-Platform:~$ cat /sys/fs/cgroup/system.slice/docker-1cd6a68cd2da420ca5764ffd6ad63d5780a78708384554233cfb94e78b48d4f3.scope/cp
u.max
100000 100000
aiysha@aiysha-VMware-Virtual-Platform:~$ cat /sys/fs/cgroup/system.slice/docker-1cd6a68cd2da420ca5764ffd6ad63d5780a78708384554233cfb94e78b48d4f3.scope/me
mory.max
73400320
Show Apps
aiysha@aiysha-VMware-Virtual-Platform:~$ S
```

→At cpu.max

It was specified that apache will be limited to 1 core, so what does 100000 100000 represents?

quota = 100000 µs (microseconds of CPU time allowed per period)

period = 100000 µs (the length of the scheduling period).

Quota ÷ Period = 1 → where the container quota = the whole length of the scheduling period, this means it can utilize the 1 cpu core.

→At the memory: 70MB = 70 × 1024 × 1024 = 73400320

Bonus task:

- First, what is OOM killer?
  in our case, Cgroup enforces memory limit to the container. So, when a process inside the container tries to exceed this limit, the kernel OOM killer terminates the process or the whole container.
  Case of process only: when the container is running more than one process it will only kill that consuming the resources by assigning **OOM score** and the process with the highest score is killed first.
  Case of whole container: If the container has only **one main process** (like Apache or Nginx), and that process is killed, the container stops entirely. Killing process with PID = 1, means the whole container terminates.

- Run an alpine container called memory-eater with memory limit of 50M:
  then exec bash:

```
aiysha@aiysha-VMware-Virtual-Platform:~$ sudo docker run -d --name memory-eater --memory="50m" --memory-swap="50m" alpine sleep infinity
[sudo: authenticate] Password:
Unable to find image 'alpine:latest' locally
latest: Pulling from library/alpine
9e595aac14e0: Download complete
aa817ad3aea: Download complete
Digest: sha256:25109184c71bdad752c8312a8623239686a9a2071e8825f20acb8f2198c3f659
Status: Downloaded newer image for alpine:latest
e03e686cb12f4b75e7d025a9914e0abed5cda40dfab4dcc3d6b1358f791c995f
aiysha@aiysha-VMware-Virtual-Platform:~$ sudo docker exec -it memory-eater sh
```

  →memory=50m : This sets the **hard limit of physical RAM** the container can use.
  →memory-swap=50m: This sets the **total memory available = RAM + swap**.
  This will strict the container to 50MB total and will can't use swap.

- At installing the stress-ng package:

```
/ # apk add stress-ng
(1/5) Installing judy (1.0.5-r1)
  0%                                                                    Killed
/ # ps
PID   USER     TIME  COMMAND
    1 root      0:00 sleep infinity
    7 root      0:00 sh
   17 root      0:00 ps
/ #
```

  →Installing stress-ng pulls in several dependencies.The package manager (apk) needs to **download, unpack, and temporarily hold files in memory** while installing.
  That memory usage can easily spike above **50 MB**,.The kernel's **OOM killer** saw the container exceed its cgroup memory limit and killed the process immediately.
  →ps shows PID = 1 still working so that container itself wasn't killed.

- Another way by forcing the command to buffer ~100 MB of data.

```
/ # tail -c 104857600 /dev/zero > /dev/null
Killed
/ #
```

--100MB = 100 * 1024 * 1024

- Checking logs:

```
aiysha@aiysha-VMware-Virtual-Platform:~$ cat /sys/fs/cgroup/system.slice/docker-e03e686cb12f4b75e7d025a9914e0abed5cda40dfab4dcc3d6b1358f791c995f.scope/me
mory.events
low 0
high 0
max 187845
oom 3
oom_kill 2
oom_group_kill 0
aiysha@aiysha-VMware-Virtual-Platform:~$
```

- To observe the whole container is killed, we will run another one and we will run tail command directly as the main process:

```
aiysha@aiysha-VMware-Virtual-Platform:~$ sudo docker run --name memory-eater-2 --memory="50m" --memory-swap="50m" alpine tail -c 104857600 /dev/zero > /d
ev/null
aiysha@aiysha-VMware-Virtual-Platform:~$ sudo docker ps -a
CONTAINER ID   IMAGE          COMMAND                  CREATED          STATUS                      PORTS                                      NAMES
afb70ebfd48c   alpine         "tail -c 104857600 /…"   12 seconds ago   Exited (137) 11 seconds ago                                            memory-ea
ter-2
e03e686cb12f   alpine         "sleep infinity"         39 minutes ago   Up 39 minutes                                                          memory-ea
ter
1cd6a68cd2da   httpd          "httpd-foreground"       31 hours ago     Up 31 hours                 0.0.0.0:8081->80/tcp, [::]:8081->80/tcp    apache-IT
I
4b686eeb9659   nginx:alpine   "/docker-entrypoint.…"   32 hours ago     Up 32 hours                 0.0.0.0:8080->80/tcp, [::]:8080->80/tcp    name-ITI
05b01e6dedbd   hello-world    "/hello"                 32 hours ago     Exited (0) 32 hours ago                                                quirky_sh
annon
aiysha@aiysha-VMware-Virtual-Platform:~$ sudo docker exec -it memory-eater-2 sh
Error response from daemon: container afb70ebfd48c25f04e6992329cccb5129ffb8a999862943aa7bfc0b5687200b6 is not running
aiysha@aiysha-VMware-Virtual-Platform:~$
```

➔ The whole container is terminated status Exited 137
➔ When trying to open bash →the container is not running